

49. In human body, which one of the following hormones regulates blood calcium and phosphate ?

- (a) Glucagon (b) Growth hormone
(c) Parathyroid hormone (d) Thyroxine

I.A.S. (Pre) 2007

Ans. (c)

Parathyroid hormone (PTH), secreted by the chief cells of the parathyroid glands, is a polypeptide containing 84 amino acids. Parathyroid hormone regulates serum calcium through its effects on the tissues. PTH reduces the reabsorption of phosphate from the proximal tubule of the kidney, which means more phosphate is excreted through in the urine.

50. Consider the following statements :

Assertion (A) : During sleep, bright light is undesirable.

Reason (R) : In bright light, production of melatonin in the body is considerably increased.

Select your answer using the codes given below.

Code :

- (a) Both (A) and (R) are correct, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are correct, but (R) is not correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.R.O./A.R.O. (Pre) 2014

Ans. (c)

Melatonin is a hormone secreted by the pineal gland in the brain. It helps to regulate other hormones and maintains the body's circadian rhythm. The circadian rhythm is an internal 24-hour 'clock' that plays a critical role at the time when we fall asleep and when we wake up. When it is dark, our body produces more melatonin. When it is bright, the production of melatonin drops. Clearly, the reason (R) is false, but assertion (A) is correct.

51. Pheromones are found in –

- (a) Insects (b) Snakes
(c) Birds (d) Bats

R.A.S./R.T.S. (Pre) 2008

Ans. (a)

Pheromones are chemical substances produced and released into the environment by an animal especially insect, affecting the behavior or physiology of others of its species.

52. Pheromones are generally produced by –

- (a) Snakes (b) Birds
(c) Bats (d) Insects

U.P.P.C.S. (Pre) 2015

Ans. (d)

See the explanation of above question.

IX. Reproduction and Embryo Development

Notes

Reproduction :

- **Reproduction** is the biological process by which new individual organisms – 'offspring' – are produced from their parents.
- Reproduction is a fundamental feature of all known life ; each individual organism exists as the result of reproduction.
- There are two forms of reproduction : **asexual** and **sexual**.
- In **asexual reproduction**, an organism can reproduce without the involvement of another organism.
- By asexual reproduction, an organism creates a genetically **similar** or **identical copy** of itself.
- Asexual reproduction is not limited to single-celled organisms. The **cloning** of an organism is a form of asexual reproduction.
- **Sexual reproduction** typically requires the sexual interaction of two specialized organisms, called **gametes**, which contains half the number of **chromosomes** of normal cells and are formed by **meiosis**, with typically a male fertilizing a female of the same species to create a fertilized **zygote**.
- This produces **offspring** organisms whose genetic characteristics are derived from those of the two parental organisms.

Human Reproduction :

- Reproduction in human beings is by sexual reproduction where both the male and female gametes fertilize to give rise to an embryo.
- In human, fertilization (fusion of male gamete – **sperms** with female gamete – **ovum**) occurs inside the body of female i.e. in the **oviduct** or **fallopian tube** hence this fertilization is known as **internal fertilization**.
- Human beings are **viviparous**. It means that they give birth to a child.

Male reproductive organs :

- **Testes** : The testes are the primary organs of male reproduction. They are situated in a bag of skin known **scrotum**. It lies outside the **abdomen cavity**.
- Testes are responsible for producing the male gamete known as **sperm** by a process called as spermatogenesis. This takes place in **seminiferous tubules** of the testes.

- **Leydig cells** of these testes synthesize testosterone. The hormone testosterone not only helps in the formation of the sperm but also in the development of secondary sexual characters in males such as deepening of the voice and development of facial and pubic hair during puberty.
- **Sperm Duct** : These are thin ducts that arise from the testes.
- **Prostate Gland** : This gland is an accessory exocrine gland. It pours its secretions into the sperm duct along with the seminal fluid from the testes and into the penis.
- **Urethra** : It functions as a common pathway for the seminal fluid and urine in males. It is longer in males than females.
- **Penis** : It is an external intromittent organ that functions to discharge both urine and semen.

Female reproductive organs :

- The female reproductive organs in human are entirely within the abdominal cavity. It comprises of the following structures :
- **Ovaries** : They give rise to the female gamete or the egg. Each ovary releases one egg (**ovum**) alternatively each month. **Ovulation** is the process of release of an egg from either one of the ovaries.
- **Fallopian tubes** : It is also known as the **oviducts**. These tubes arise from the ovaries and end at the **uterine fundus**. Their function is to pick up the egg and usually fertilization takes place in it.
- **Uterus** : It is a large muscular organ. It is the region of the development of the **fetus**.
- **Cervix** : It is a cylindrical muscle ring that is present towards the lower end of the uterus that dilates at the time of delivery of the baby.
- **Vagina** : It is a muscular tube-like structure that is present at the lower end of the cervix and leads towards the outside of the female body. The vagina functions as the pathway for the penis to enter the female body and deposit the sperms.
- **Urethra** : In females, urethra serves only for the passage of urine.

Fertilization and Embryo Development :

- The egg is formed under the influence of both female sex hormones estrogen and progesterone. This process is known as **Oogenesis**.
- The fertilization capacity of sperm is about 48 hours to 72 hours while egg's capacity is about 12 hours to 24 hours after ovulation.
- Both the egg and sperm **fertilize in oviduct** and form the **diploid zygote** which contains 23-23 chromosomes from each parent.
- After fertilization, **zona pellucida** from the egg forms a thick layer around the zygote to prevent more than one sperm to fertilize the egg.
- Zygote now turns into a **morula** and then into a **blastocyst**. It then develops **chorionic villi** from the outer layer of the blastocyst known as the **chorion**.
- These **villi** attach themselves to the inner wall of the uterus. **Implantation** is the process of attachment of the embryo to the uterine wall. The tissue between the growing embryo and the mother's uterine wall form the **placenta**.
- Placenta functions to provide nutrition to the embryo until its birth.
- The hormones estrogen and progesterone both help in maintaining the placenta and the fetus inside the uterus.
- The normal gestation period for human is about 38 weeks which is a little over 9 months.
- At the end of this term, the uterine contractions begins under the influence of **oxytocin** hormone. It affects the cervix and causes it to dilate to allow the baby to pass outside the body of the mother.
- The **umbilical cord** with its blood vessel and the placenta are also expelled along with the baby.
- Prenatal stages and development include **germinal**, **embryonic** and **fetal** period.
- The first two weeks of development is the **germinal period**, which begins with conception and ends when the blastocyst is fully implanted into uterine tissue.
- **Embryonic period** of prenatal development lasts from implantation until about 8 weeks from the time of conception (or the 10th week of pregnancy). This is the most important time of prenatal development because the embryo is developing the foundations for a healthy baby.
- Three germ layers are formed in this period – The outer layer is called the **ectoderm**, the middle layer is called the **mesoderm** and inner layer is called the **endoderm**.
- Each germ layer is differentiated into different structures. The **ectoderm** forms many outer tissues such as skin, hair, nervous system tissues including brain. The **mesoderm** forms tissues inside the body such as lungs, bones and muscles including the heart. The **endoderm** forms tissues such as the digestive tract and bladder as well as other internal organs.

- **Fetal period :** Once cell differentiation is mostly complete, the embryo enters the fetal period and becomes known as a fetus. This period of development begins during the ninth week and lasts until birth. In this period the formed organs are mature and growth takes place.

The heart of fetus starts to beat in 4th-5th week of its development.

- **Amnion :** The amnion is a membrane that closely covers the embryo when it first formed. It is filled with **amniotic fluid**, which causes the amnion to expand and become the amniotic sac, which serves to provide a protective environment for the developing embryo or fetus.
- The amnion along with **chorion**, the yolk sac and the **allantois** form a protective sac around the embryo.
- Chorion regulate, the gaseous exchange while;
- Allantois helps the embryo exchange gases and handle the liquid waste.

Question Bank

1. Where does fertilization occur?

- (a) In uterus (b) In oviduct
(c) In ovary (d) In vagina

45th B.P.S.C. (Pre) 2001

Ans. (b)

Fertilization occurs at the end of the fallopian tube or oviduct away from the uterus (close to the ovary). The oviduct is the tube that links ovary to the uterus and ovulated oocyte travels down to become fertilized by sperm present in the female tract.

2. In human beings, normally in which one of the following parts, does the sperm fertilize the ovum ?

- (a) Cervix
(b) Fallopian tube
(c) Lower part of uterus
(d) Upper part of uterus

I.A.S. (Pre) 2007

Ans. (b)

In human beings, normally the sperm fertilizes the ovum in the fallopian tube. Fallopian tube, also called oviduct or uterine tube, either of a pair of long narrow ducts located in the human female abdominal cavity that transport the male sperm cells to the egg, provide a suitable environment for fertilization and transport the egg from the ovary, where it is produced.

3. After reaching the female genital tract, human sperms retain their fertilizing capacity for –

- (a) Two minutes (b) Twenty minutes
(c) Ninety minutes (d) One to two days

U.P.P.C.S. (Mains) 2010

Ans. (d)

A sperm remains alive for up to about 72 hours after entering into oviduct but its capacity to fertilize the ovum lasts for about 48 hours (2 days).

4. The heart of human embryo starts beating?

- (a) In the first week of development
(b) In the third week of its development
(c) In the fourth week of its development
(d) In the sixth week of its development

I.A.S. (Pre) 1994

Ans. (c)

About 4-5 weeks after fertilization, when the heart of human embryo first begins to beat, the sound of the little heart is too soft to hear. In 4-5 weeks, the heart of human embryo typically beats between about 105 to 121 times per minute.

5. Which one of the following is a membrane that protects the developing embryo from desiccation ?

- (a) Amnion (b) Allantois
(c) Chorion (d) Yolk sac

I.A.S. (Pre) 1995

Ans. (a)

The amnion is a membrane that covers the embryo. It is filled with the amniotic fluid which causes the amnion to expand and become the amniotic sac which serves to provide a protective environment for the developing embryo against desiccation, outer pressure etc.

6. In which organ sertoli cells are found?

- (a) Ovary (b) Kidney
(c) Liver (d) Testis

M.P. P.C.S. (Pre) 2021

Ans. (d)

Sertoli cells are somatic cells found in human testes which contribute to the process of spermatogenesis (the production of sperm) as a structural component of the seminiferous tubules. They are activated by follicle-stimulating hormone (FSH) secreted by the adenohypophysis and express FSH receptor on their membranes. They are a type of sustentacular 'nurse' cell.